

Chapter → 6

Conditional Expression

- > Sometimes we want to play PUBG on our phone if the day is Sunday.
- > Sometimes we order Ice Cream online if the day is Sunday.
- > Sometimes we go hiking if our parents allow.
- > All these are decisions which depends on a condition being met.
- > In python programming too, we must be able to execute instructions on a condition(s) being met.

★ IF ELSE AND ELIF IN PYTHON

// Code (IF ELIF ELSE LADDER)

```
a = int(input("Enter your age: "))
```

```
if (a >= 18):
```

```
    print("You are above the age of consent")
```

```
    print("Good for you")
```

```
elif (a < 0):
```

```
    print("You are entering an invalid age")
```

```
else:
```

```
    print("You are below the age of consent")
```

```
print("End of program")
```


if else and elif are multiway decision taken by our program due to certain condition in our code.

Syntax:

```
if (condition):
```

```
    print("yes")
```

```
elif (condition 2):
```

```
    print("maybe")
```

```
else:
```

```
    print("no")
```

code example:

```
a = 22
```

```
if (a > 9):
```

```
    print("greater")
```

```
else:
```

```
    print("less")
```

• Quick Quiz:

Write a program to print yes when the age entered by the user is greater than or equal to 18.

★ Relational Operators

Relational operators are used to evaluate conditions inside the if statement. Some examples of Relational operators

$==$: equals.

$>=$: greater than or equal to.

$<=$: lesser than or equal to

★ Logical Operator

In python logical operators operate on the conditions statement.

Example:

and - true if both operands are true or else false.

or - true if at least one operand is true else false.

not - inverts true to false or false to true.

★ Elif clause

elif in python means [else if] An if statements can be changed together with a lot of these elif statements followed by an else statement.

Syntax:

```
if (condition):
```

```
    # code
```

```
elif (condition 2):
```

```
    # code
```

```
elif (condition 3):
```

```
    # code
```

```
else:
```

```
    # code
```

Important Notes:

- There can be any number of elif.
- Last else is executed only if all the conditions inside elif fails.

Teacher's Signature _____

Chapter + 6

Practice Set

1. Write a Program to find the greatest of four numbers Entered by the user.
2. Write a Program whether a student has passed or failed if it requires a total of 40% and at least 33% in each subject to pass. Assume 5 subjects and take marks as an input from the user.
3. A spam comment is defined as a text containing following keywords:
"Make a lot of Money", "buy now", "Subscribe this", "click this". Write a program to detect this spams.
4. Write a Program to find whether a given username contains less than 10 characters or not.
5. Write a Program which find out whether a given name is present in list or not.
6. Write a Program to find out whether a given text is talking about "HARRY" or not.
7. Write a Program to calculate the grade of a student from his marks which are:
90-100 $\Rightarrow$ Ex, 80-90 $\Rightarrow$ A, 70-80 $\Rightarrow$ B, 60-70 $\Rightarrow$ C, 50-60 $\Rightarrow$ D, 40-50 $\Rightarrow$ E